

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 30

Recreate the CloudSLAW VPC with CloudFormation IaC

AWS CloudFormation | YAML | Amazon VPC | NAT Gateway | NACLs

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ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.0
RELEASE DATE	July 21, 2026
STATUS	Complete

Executive Summary

Logged into **TestAccount1**, authored a **YAML CloudFormation** template, created a CloudFormation stack named **CloudSlaw**, and used the stack to recreate the complete CloudSLAW VPC that was previously built manually. The template created the VPC, Internet Gateway and attachment, DHCP option set and association, two public subnets, two private subnets, public and private route tables, route table associations, public Internet Gateway route, Elastic IP, NAT Gateway, private NAT route, public and private network ACLs, NACL entries, and NACL subnet associations. After confirming the stack reached **CREATE_COMPLETE** and the VPC existed, the stack was deleted so CloudFormation removed the VPC resources.

IAC OUTCOME

This project turns a click-built network into repeatable infrastructure. The lab explicitly focuses on recreating the same VPC with CloudFormation so it can be created and destroyed on demand for cost control.

Business Problem

Projects 28 and 29 proved the VPC design manually, but manual console work is slow, inconsistent, and harder to recreate. A training environment also needs a way to avoid NAT Gateway and Elastic IP charges by destroying the network when it is not needed. CloudFormation solves both problems by defining the desired infrastructure in a template, building the resources, tracking dependencies, and deleting the stack when the lab is complete.

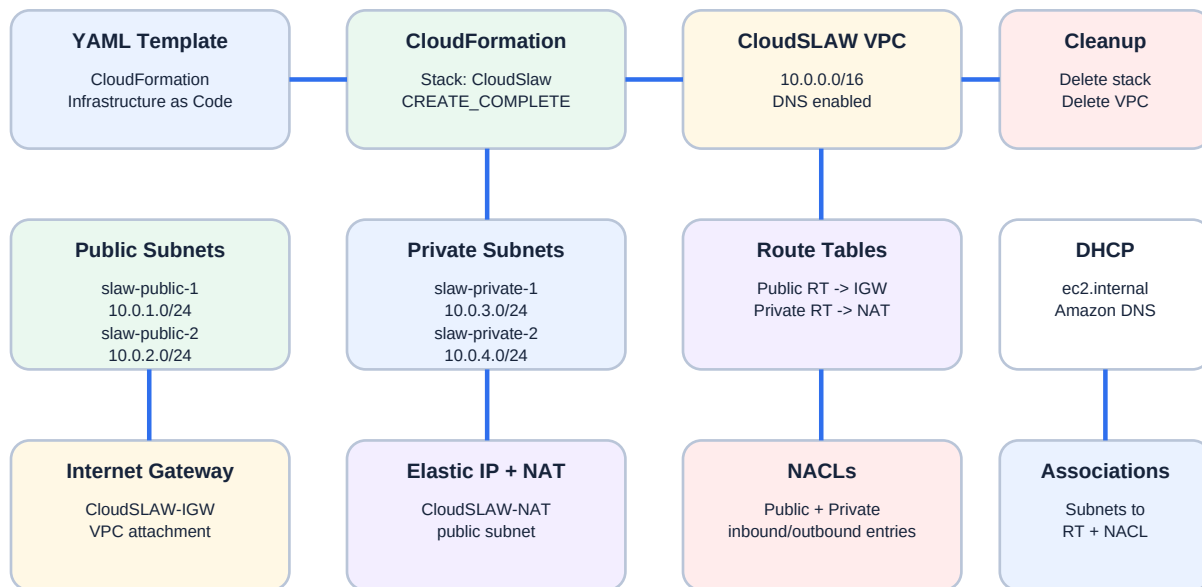
Objectives

- Use YAML to define the CloudSLAW VPC infrastructure.
- Create a VPC with CIDR 10.0.0.0/16 and DNS support/hostnames enabled.
- Create and attach an Internet Gateway.
- Create a DHCP option set, including domain name and Amazon-provided DNS, then associate it to the VPC.
- Create four subnets: two public and two private.
- Create public and private route tables.
- Associate public subnets with the public route table and private subnets with the private route table.
- Create a public default route to the Internet Gateway.
- Allocate an Elastic IP and create a NAT Gateway in the first public subnet.
- Create a private default route to the NAT Gateway.
- Create public and private NACLs, inbound/outbound entries, and subnet associations.
- Create a CloudFormation stack from the template, confirm CREATE_COMPLETE, validate the VPC, then delete the stack for cleanup.

Environment

Cloud Provider	Amazon Web Services (AWS)
Primary Service	AWS CloudFormation
Network Services	Amazon VPC, Internet Gateway, NAT Gateway, Elastic IP, Route Tables, Subnets, Network ACLs, DHCP Options
Account Context	TestAccount1 / AdministratorAccess
Region	us-west-2 / Oregon
Stack Name	CloudSlaw
Template Format	YAML
VPC Name	CloudSLAW
VPC CIDR	10.0.0.0/16
Public Subnets	slaw-public-1: 10.0.1.0/24, slaw-public-2: 10.0.2.0/24
Private Subnets	slaw-private-1: 10.0.3.0/24, slaw-private-2: 10.0.4.0/24
Routing	Public route table to IGW, private route table to NAT Gateway
Cleanup	CloudFormation stack deleted after validation

Architecture



CloudFormation defines the VPC and dependencies so the network can be recreated and deleted on demand.

Figure 1 - CloudFormation IaC deployment for the CloudSLAW VPC.

Implementation Summary

1. Signed into TestAccount1 using AdministratorAccess.
2. Created a YAML CloudFormation template with AWSTemplateFormatVersion 2010-09-09 and a CloudSLAW VPC description.
3. Defined the CloudSLAWVPC resource with CIDR 10.0.0.0/16, DNS support, DNS hostnames, and tags.
4. Defined the Internet Gateway and VPCGatewayAttachment so the gateway is attached to the VPC.
5. Defined DHCP options with ec2.internal and AmazonProvidedDNS, then associated those options to the VPC.
6. Defined two public subnets and two private subnets across the first two available Availability Zones.
7. Defined public and private route tables and route table associations for each subnet.
8. Defined the public route so 0.0.0.0/0 uses the Internet Gateway.
9. Defined an Elastic IP and NAT Gateway in the first public subnet.
10. Defined the private route so private-subnet 0.0.0.0/0 traffic targets the NAT Gateway.
11. Defined public and private NACLs, inbound/outbound entries, and NACL subnet associations.
12. Uploaded the YAML template to CloudFormation and created the CloudSlaw stack.
13. Confirmed the stack reached CREATE_COMPLETE.

14. Validated the created VPC in the VPC resource map.

15. Deleted the CloudFormation stack so CloudFormation removed the VPC infrastructure.

CloudFormation Resource Map

VPC core	CloudSLAWVPC, CloudSLAWDHCPOptionSet, CloudSLAWVPCDHCPOption sAssociation	Creates the network boundary and required DNS/DHCP behavior.
Internet access	CloudSLAWInternetGateway, CloudSLAWAttachGateway, PublicRoute	Connects public subnets to the Internet Gateway.
Subnets	SlawPublicSubnet1/2, SlawPrivateSubnet1/2	Creates public and private address ranges across two AZs.
Routing	PublicRouteTable, PrivateRouteTable, subnet route table associations	Separates public and private network paths.
NAT egress	CloudSLAWElasticIP, CloudSLAWNATGateway, PrivateRoute	Allows private subnet outbound Internet traffic through NAT.
NACLs	PublicNetworkAcl, PrivateNetworkAcl, entries and associations	Defines subnet-level network ACL resources and associations.

Template Excerpt

```
AWSTemplateFormatVersion: '2010-09-09'
Description: CloudSLAW CloudFormation template to create a VPC with public and private subnets, and a NAT gateway

Resources:
  # VPC
  CloudSLAWVPC:
    Type: 'AWS::EC2::VPC'
    Properties:
      CidrBlock: '10.0.0.0/16'
      EnableDnsSupport: 'true'
      EnableDnsHostnames: 'true'
      Tags:
        - Key: Name
          Value: CloudSLAW

  # Internet Gateway
  CloudSLAWInternetGateway:
    Type: 'AWS::EC2::InternetGateway'
    Properties:
      Tags:
        - Key: Name
          Value: CloudSLAW-IGW

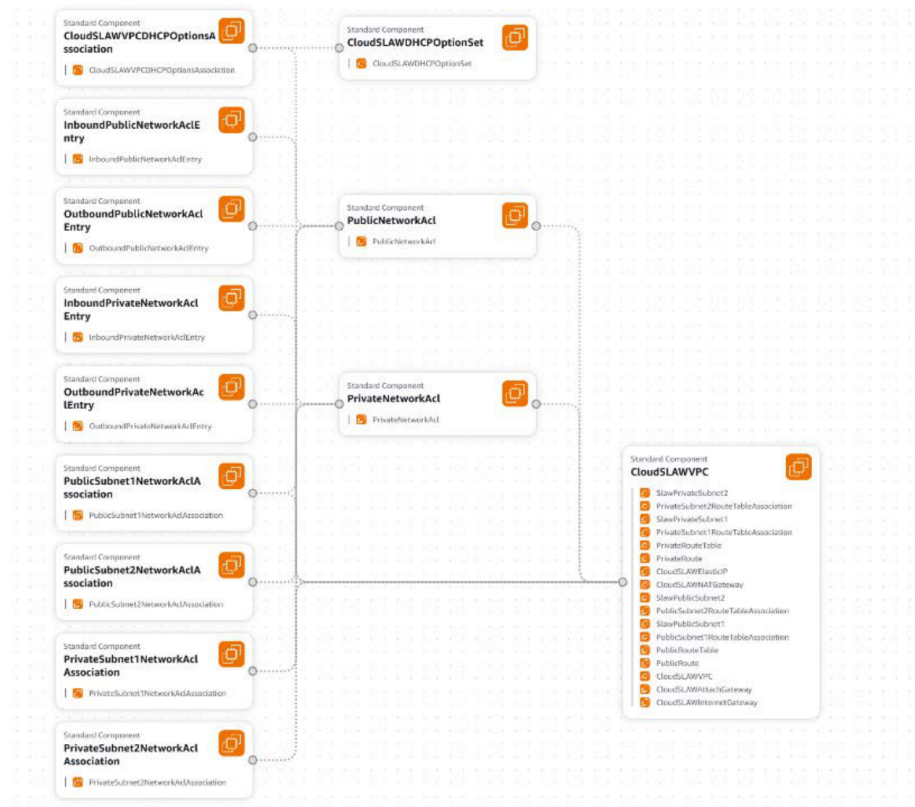
  CloudSLAWAttachGateway:
    Type: 'AWS::EC2::VPCGatewayAttachment'
    Properties:
      VpcId: !Ref CloudSLAWVPC
      InternetGatewayId: !Ref CloudSLAWInternetGateway

  # DHCP Options Set
  CloudSLAWDHCPOptionSet:
    Type: 'AWS::EC2::DHCPOptions'
    Properties:
      DomainName: 'ec2.internal'
      DomainNameServers:
        - 'AmazonProvidedDNS'
      Tags:
        - Key: Name
          Value: CloudSLAW-DHCP

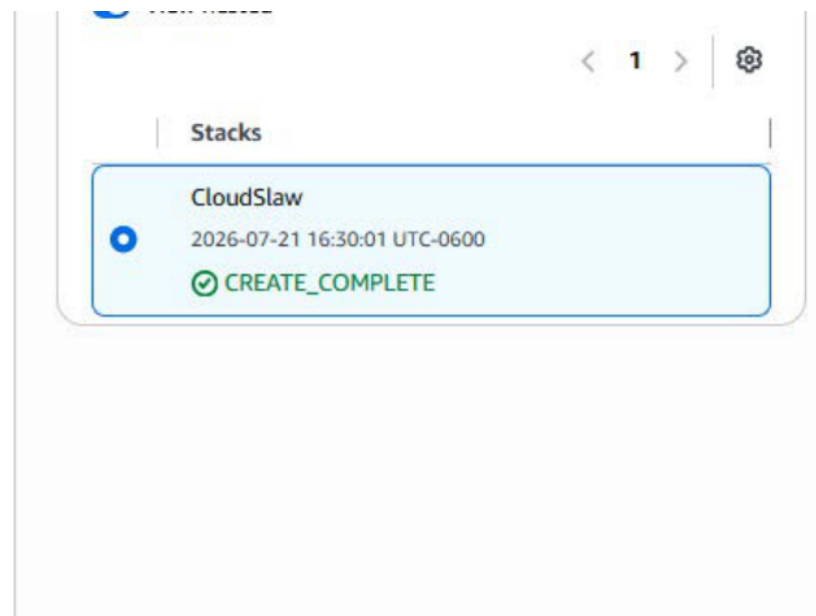
  CloudSLAWVPCDHCPOptionsAssociation:
    Type: 'AWS::EC2::VPCDHCPOptionsAssociation'
```

Excerpt from the submitted YAML template showing the template header, VPC resource, Internet Gateway, VPC Gateway Attachment, and DHCP option set.

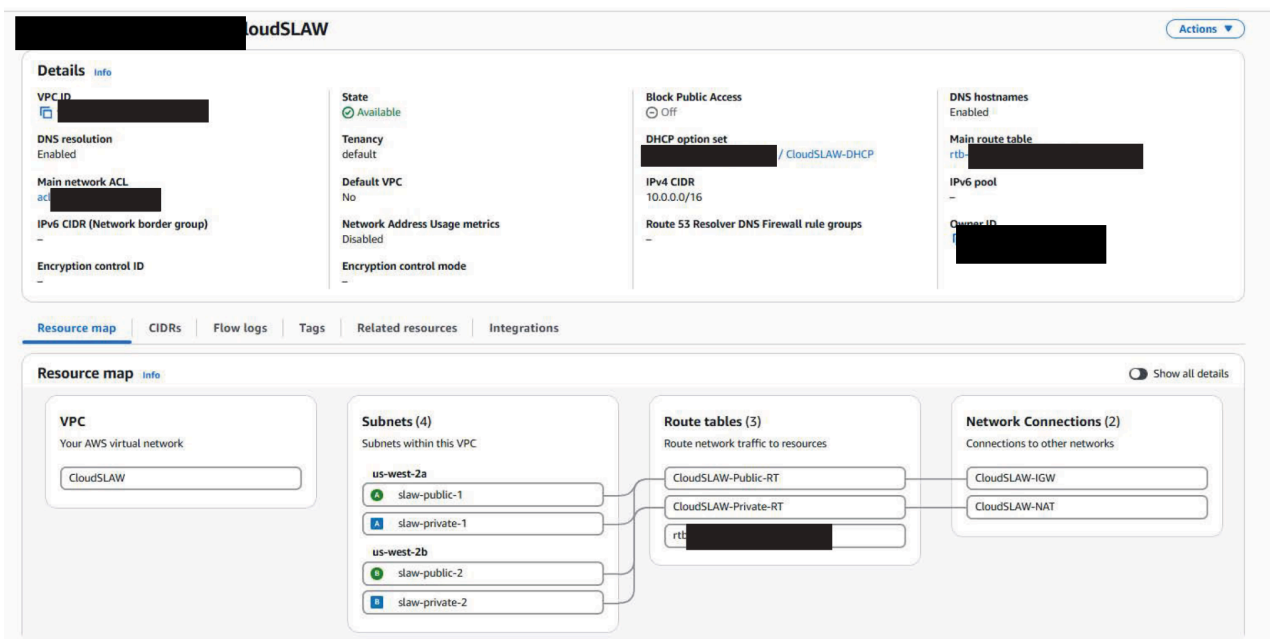
Evidence

Figure 2 - CloudFormation Dependency Flowchart

The CloudFormation flowchart evidence shows standard components and dependencies, including DHCP option association, DHCP option set, public and private network ACL entries, network ACL resources, subnet NACL associations, and the CloudSLAWVPC grouping that includes subnets, routes, NAT Gateway, Elastic IP, Internet Gateway, and route table associations.

Figure 3 - CloudFormation Stack Created Successfully

The stack evidence shows stack CloudSlaw with status CREATE_COMPLETE, confirming CloudFormation successfully created the infrastructure from the YAML template.

Figure 4 - Final VPC Resource Map

The final VPC evidence shows CloudSLAW available with IPv4 CIDR 10.0.0.0/16, DNS resolution and hostnames enabled, CloudSLAW-DHCP attached, four subnets across us-west-2a and us-west-2b, CloudSLAW-Public-RT, CloudSLAW-Private-RT, CloudSLAW-IGW, and CloudSLAW-NAT.

Security Analysis

This project moves the VPC design from manual console operations into repeatable Infrastructure as Code. That improves consistency and makes the design reviewable in source control.

The public/private subnet split is preserved from the manual labs: public subnets route to the Internet Gateway, while private subnets route outbound Internet traffic to the NAT Gateway.

The template uses explicit route table associations so subnet behavior is deterministic rather than relying on the default route table.

The NACLs are included to make the VPC complete, but the lab's NACL rules are permissive. In production, network ACLs should be intentionally designed or kept simple while stronger control is applied with security groups and route architecture.

Deleting the CloudFormation stack is a cost-control practice and a validation of IaC reversibility. If the stack owns the resources, stack deletion should remove them cleanly.

Lessons Learned

CloudFormation resources use logical IDs such as CloudSLAWVPC that can be referenced by other resources in the same template.

!Ref lets one part of the template reference IDs that CloudFormation creates during deployment.

CloudFormation can determine many resource dependencies, but explicit dependencies such as DependsOn can improve correct sequencing and readability.

Infrastructure as Code makes the network easier to recreate, review, version, and delete compared to click-by-click console work.

A complete VPC design requires more than the VPC object: subnets, route tables, routes, gateway attachments, DHCP behavior, NAT, and associations all matter.

Production Improvements

- Move the template into GitHub and version it with the portfolio project.
- Add Parameters for CIDR ranges, tags, and environment names.
- Add Outputs for VPC ID, subnet IDs, route table IDs, NAT Gateway ID, and Internet Gateway ID.
- Use Conditions to optionally deploy NAT Gateway for cost-controlled lab runs.
- Use one NAT Gateway per Availability Zone for production high availability.
- Add VPC Flow Logs and log destination configuration.
- Add VPC endpoints for S3 and Systems Manager to reduce NAT dependency and cost.
- Validate the template with cfn-lint and CloudFormation change sets before deployment.

Skills Demonstrated

AWS CloudFormation	YAML	Amazon VPC
Subnet architecture	Route tables	Internet Gateway / NAT Gateway
NACL associations	Elastic IP	Stack lifecycle management

Resume Bullet

RESUME

Recreated a multi-AZ AWS VPC using CloudFormation YAML, including public and private subnets, Internet Gateway, NAT Gateway, route tables, DHCP options, NACLs, subnet associations, stack validation, and stack-based cleanup.

STAR Interview Story

Situation: The VPC had been built manually and then deleted to avoid NAT Gateway charges, but the environment needed a repeatable way to recreate it.

Task: Build the same VPC design using CloudFormation and validate that it could be created and removed as a stack.

Action: I wrote a YAML template defining the VPC, gateways, DHCP options, four subnets, route tables, routes, NAT, NACLs, and associations, created the CloudSlaw stack, confirmed CREATE_COMPLETE, validated the VPC, and deleted the stack.

Result: The lab now has a repeatable Infrastructure as Code pattern for creating and destroying the CloudSLAW VPC on demand.

Version History

v1.0.0	July 21, 2026	Initial documented implementation. Created CloudFormation YAML for the full CloudSLAW VPC, created and validated the CloudSlaw stack, confirmed final VPC resources, deleted the stack for cleanup, and embedded provided evidence using portfolio documentation standard v1.0.1.

References

- Cloud Security Lab a Week (S.L.A.W.) - Build a VPC with IaC and Maybe TLC.
- User-provided YAML VPC template.
- User-provided CloudFormation flowchart screenshot.
- User-provided CloudFormation stack completion screenshot.
- User-provided final VPC resource map screenshot.
- Project 28 - Custom VPC Public Subnets and Internet Gateway Routing.
- Project 29 - Private Subnets and NAT Gateway Egress.